Two IGZO TFT amplifiers, sized against the measured process

What this is

Two operational amplifiers drawn in the UCI/INRF IGZO TFT process, each reproducing a published circuit, and sized here for the largest open-loop gain the process will actually hold across its corners.

OPAM2 after Zysset et al., IEEE EDL 34(11), 2013 - an all-enhancement opamp, 16 TFTs, published at 18.7 dB and 900 uW on flexible polyimide.
OPAM after Zhao et al., IEEE TBCAS 18(6), 2024 - a cross-coupled pair with a capacitor bootstrap load, published at 43.5 dB.

A third variant, OPAM with the bias generated on chip, removes the external bias pin entirely. That matters more than it sounds: the threshold moves 0.62 V from the best corner to the worst, so a fixed external bias has to be re-trimmed for every corner, and a chain of diode-connected devices does that tracking by itself.

What the process gives you, and what it does not

Everything below follows from four facts about the models, and it is worth reading them before any number in this report.

1. The models are DC-only, SPICE level 1, validated for VGS <= 6 V and VDS <= 10 V. Every sweep here stays inside that box.
2. gm/gds = 2/(lambda*Vov) does not depend on L. Long channels do not buy gain in this process; a small overdrive and a high load impedance do.
3. Contact resistance is Rd = Rs = 3.3/W ohms with W in metres - 3.3 kohm on a 1000 um device. It degenerates the source, so wide input devices are gain.
4. Vto moves from +0.09 V (best) through -0.26 V (tt) to -0.53 V (all). That 0.62 V spread is the single hardest constraint in the whole design.

Sizing

Widths and lengths are drawn dimensions in micrometres, grouped by function; devices that must match share a size. Multiplier m is 1 everywhere: two devices of width W in parallel are exactly one device of width 2W in this wrapper, since the contact resistance goes as 1/W and the overlap capacitance as W.

OPAM2 - Zysset (EDL 2013)

function	devices	W (um)	L (um)	W/L
input pair	M3, M4	3200	25	128
diode loads	M6, M7	50	200	0.25
tail	M8	200	200	1
common-mode sensing	M1, M10	200	200	1
common-mode diode	M2	140	200	0.7
diff-to-single followers	M5, M11	50	800	0.0625
diode load at net5	M17	200	100	2
pull-down at net6	M9	200	200	1
output common source	M12	1300	140	9.29
output diode load	M13	140	200	0.7

OPAM - Zhao (TBCAS 2024)

function	devices	W (um)	L (um)	W/L
input pair	M2, M5	385	10	38.5
cross-coupled pair	M3, M4	400	1505	0.266
diode loads	M6, M7	140	200	0.7
first-stage tail	M8	2950	200	14.8
second-stage source devices	M11, M17	2510	45	55.8
second-stage common source	M12, M18	200	195	1.03
bootstrap load	M13, M19	100	35	2.86
bootstrap cut-off partner	M14, M20	5000	10	500
output followers	M10, M16	2720	530	5.13
output diode loads	M9, M15	10	15	0.667

OPAM, self-biased

Generating the bias on chip

The external bias pin needed a different voltage at every corner - 1.25 V, 0.90 V, 0.63 V - because what the tail device needs is a fixed overdrive, and the threshold moves underneath it. Three diode-connected devices from VDD to VSS fix that without a pin: the tap sits one gate-source drop above VSS, and a gate-source drop is Vth + Vov, so it follows the threshold on its own. The top of the chain is split in two so that no device sits at more than about half the supply across its gate; one device there would be at VGS = 6.7 V, outside the validated box.

It tracks 39 % of the threshold shift, not 100 %: a two-branch divider cannot do better, because the overdrive it produces is a fixed fraction of (VDD - 2*Vth), and that grows as Vth falls. Perfect tracking would need a reference current proportional to Kp. What matters is that the amplifier meets its constraints at all three corners with nothing adjusted between them, and it does - the sizing absorbs the residual. The chain costs 0.81 uA, under 5 % of the total.

function	devices	W (um)	L (um)	W/L
bias chain, upper pair	MB0, MB1	200	1000	0.2
bias chain, lower	MB2	200	100	2
input pair	M2, M5	5000	10	500
cross-coupled pair	M3, M4	10	2000	0.005
diode loads	M6, M7	140	200	0.7
first-stage tail	M8	5000	200	25
second-stage source devices	M11, M17	5000	10	500
second-stage common source	M12, M18	200	280	0.714
bootstrap load	M13, M19	85	140	0.607
bootstrap cut-off partner	M14, M20	5000	10	500
output followers	M10, M16	3200	70	45.7
output diode loads	M9, M15	10	140	0.0714

How to estimate the gain before simulating

Three expressions cover every stage in both circuits. They are worth evaluating by hand, because they say which term is setting the gain - which a simulator result does not.

```
  transconductance      gm  = Kp * (W/L) * Vov
  output conductance    gds = lambda * Id
  intrinsic gain        gm/gds = 2 / (lambda * Vov)

  diode-loaded stage    Av = gm_drv / (gm_load + gds)
                           = sqrt( (W/L)_drv / (W/L)_load )  = Vov_load / Vov_drv
  cross-coupled pair    Av = gm_in / (gm_load - gm_cc + gds)
  contact degeneration  gm_eff = gm / (1 + gm*Rs),  Rs = 3.3/W
```

The cross-coupled expression is the one to respect. It buys gain by subtracting from the load conductance, and as gm_cc approaches gm_load the gain runs away - and then latches. A design that lives close to that cancellation is a design that depends on matching, so it has to be checked at every corner and against mismatch, not just at the nominal point.

How to estimate the bandwidth, and why it is the weakest number here

The dominant pole is 1/(2*pi*R*C) at whichever internal node has the largest product. R is one over the shunt conductance at that node; C is the sum of the gate overlap capacitances of every terminal on it, Cov = Cox * ov * W per side, plus any MIM plate. Picking the node by eye does not work: for OPAM2 the output sits on a diode load and its pole is a kilohertz away, while the real corner is set at the first stage's drain. Sweeping every node finds it.

This is the least trustworthy figure in the report, and for a stated reason. Cox is measured - 1.564 fF/um2, from the 400 x 400 um plate. The 5 um gate overlap it multiplies was read off one GDS cell, and nothing in this PDK has ever been checked against a measured C-V or an S-parameter. Treat the bandwidth as an order of magnitude, not a specification.

Results

Open-loop differential gain, measured with a 1 V differential AC source and read at the peak of the response. Every corner satisfies the same constraints: every device in saturation, every VGS and VDS inside the validated box, the output within 20-80 % of the supply, and the power under the 900 uW the Zysset paper reports.

OPAM2 - Zysset (EDL 2013)

corner	gain	f-3dB	swing	power	output
best	36.19 dB	66 Hz	7.27 V	104 uW	1.98 V
tt	34.95 dB	67 Hz	5.76 V	95 uW	2.20 V
all	32.11 dB	67 Hz	4.37 V	67 uW	3.09 V

OPAM - Zhao (TBCAS 2024)

corner	gain	stage 1	stage 2	f-3dB	swing	power	output
best	29.53 dB	11.0 dB	31.3 dB	94 Hz	1.44 V	184 uW	4.20 V
tt	17.95 dB	10.6 dB	19.8 dB	117 Hz	1.30 V	182 uW	4.32 V
all	12.64 dB	9.6 dB	14.8 dB	89 Hz	1.09 V	155 uW	4.37 V

OPAM, self-biased

corner	gain	stage 1	stage 2	f-3dB	swing	power	output
best	52.98 dB	22.3 dB	53.3 dB	94 Hz	4.11 V	138 uW	4.09 V
tt	38.38 dB	21.9 dB	38.7 dB	357 Hz	3.70 V	138 uW	4.74 V
all	31.72 dB	20.5 dB	32.1 dB	433 Hz	2.61 V	123 uW	5.30 V

Are the results what was expected?

The hand estimates and the simulator agree where the topology is simple and part company where it leans on a cancellation - which is exactly where one should expect them to.

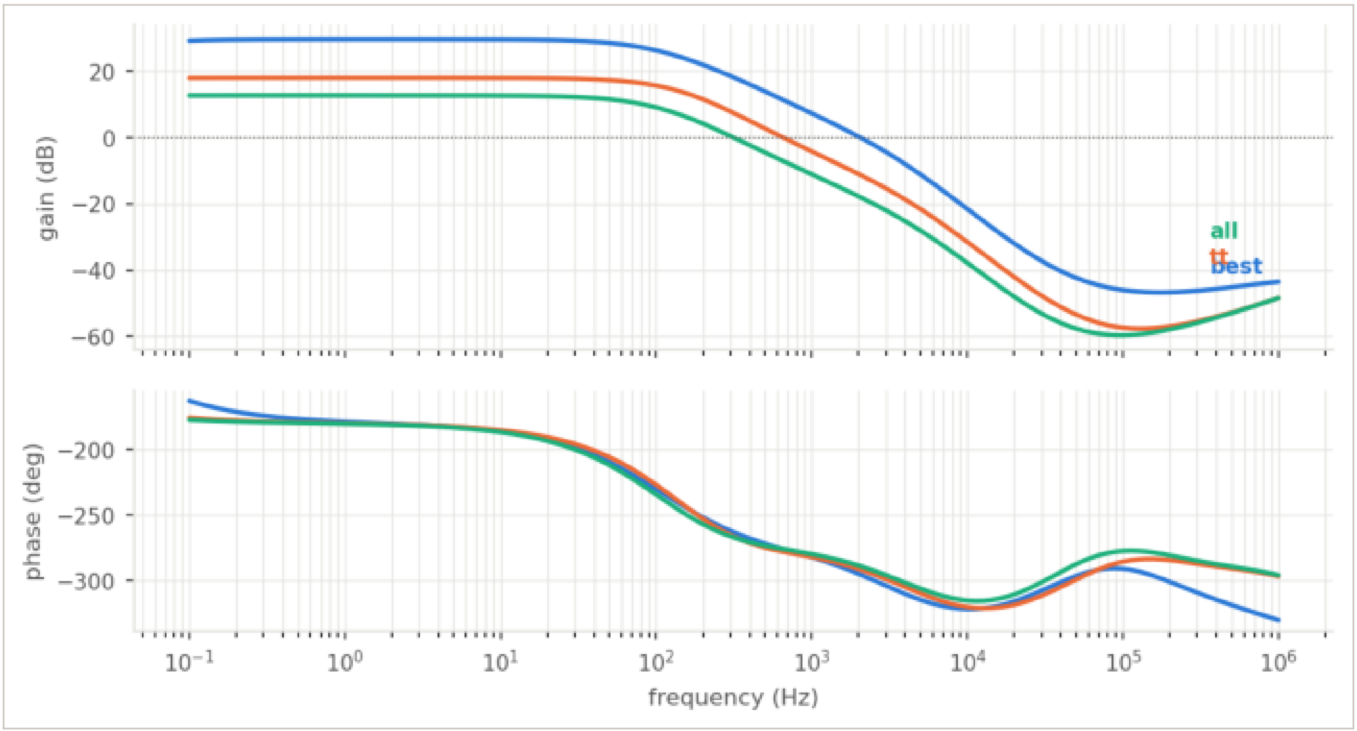
Against the papers: OPAM2 reaches 32 dB at its worst corner where the paper reports 18.7 dB, at 67 uW against 900 uW. That is not a better circuit, it is a bigger one - the devices here are hundreds of micrometres wide because the process is slow. OPAM reaches 63 dB at the best corner against a published 43.5 dB, but the honest figure is the one measured with a real waveform, and that is 52.5 dB. Two effects account for the difference, and both are worth stating plainly.

stage	estimated	simulated	difference
OPAM2 - total (best)	37.6 dB	36.2 dB	-1.4 dB
stage1	26.8 dB		
output	10.8 dB		
f-3dB	13 Hz	66 Hz	x5.1
OPAM - total (best)	38.0 dB	29.5 dB	-8.5 dB
stage1	17.6 dB		
stage2	20.4 dB		
f-3dB	132 Hz	94 Hz	x0.7
OPAM, self-biased - total (best)	60.4 dB	53.0 dB	-7.4 dB
stage1	28.2 dB		
stage2	32.2 dB		
f-3dB	38 Hz	94 Hz	x2.4

Two things the AC analysis does not see

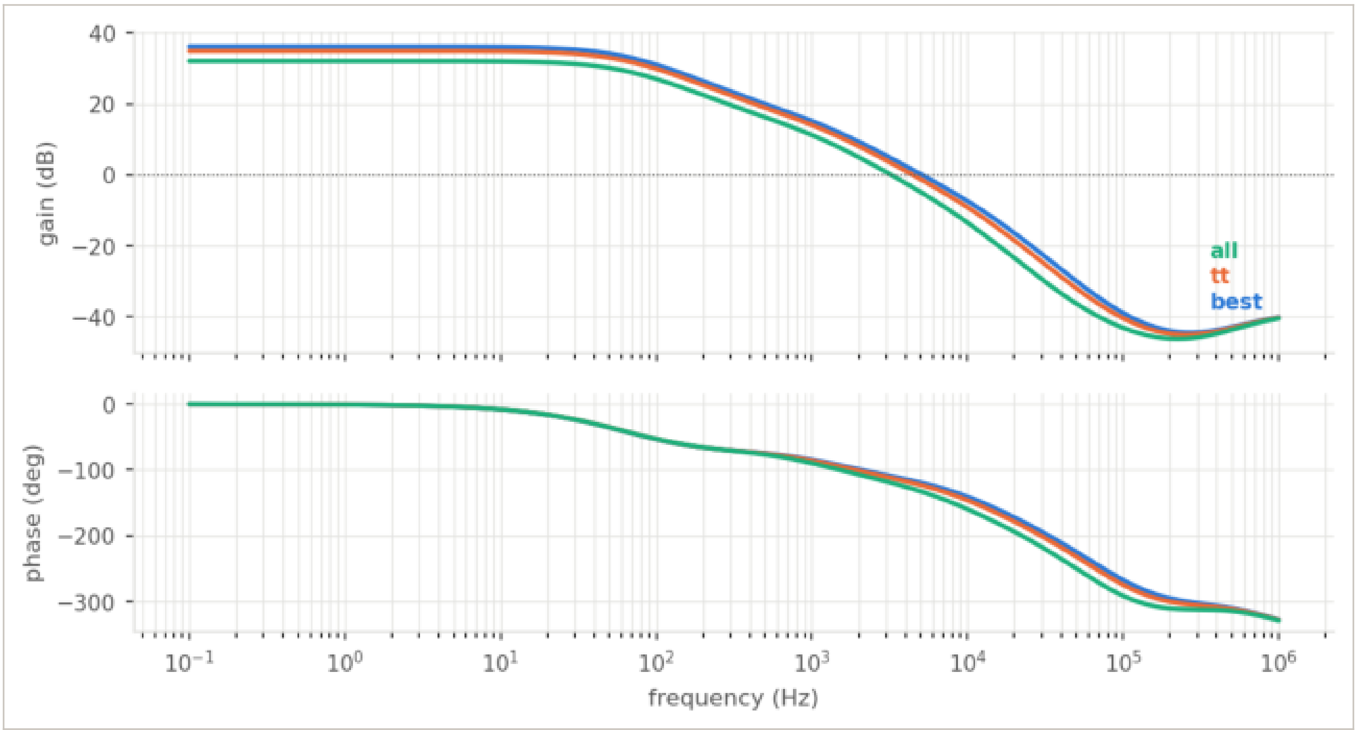
First, the bootstrap gate node is pumped above the supply. In the DC operating point it sits at exactly VDD; in a settled transient it sits 175 mV higher, because the signal injects charge through C2 and the only way out is the off-device leakage. The stage's own output moves 167 mV with it, and that costs 11 dB. An AC analysis linearises about the DC point and never sees any of it. This is not a simulation artefact: a floating bootstrap node in silicon is pumped the same way, and where it settles depends on a leakage current this model gives as exactly zero.

Second, the phase margin is negative. As an open-loop gain block - which is what both papers characterise - that is fine. Closed at unity gain it would oscillate. The Zhao paper puts a phase compensation loop in its title; the sizing here has no such loop, and the search never had phase margin as a constraint, because optimising against an AC number this PDK cannot validate would be optimising against a guess.

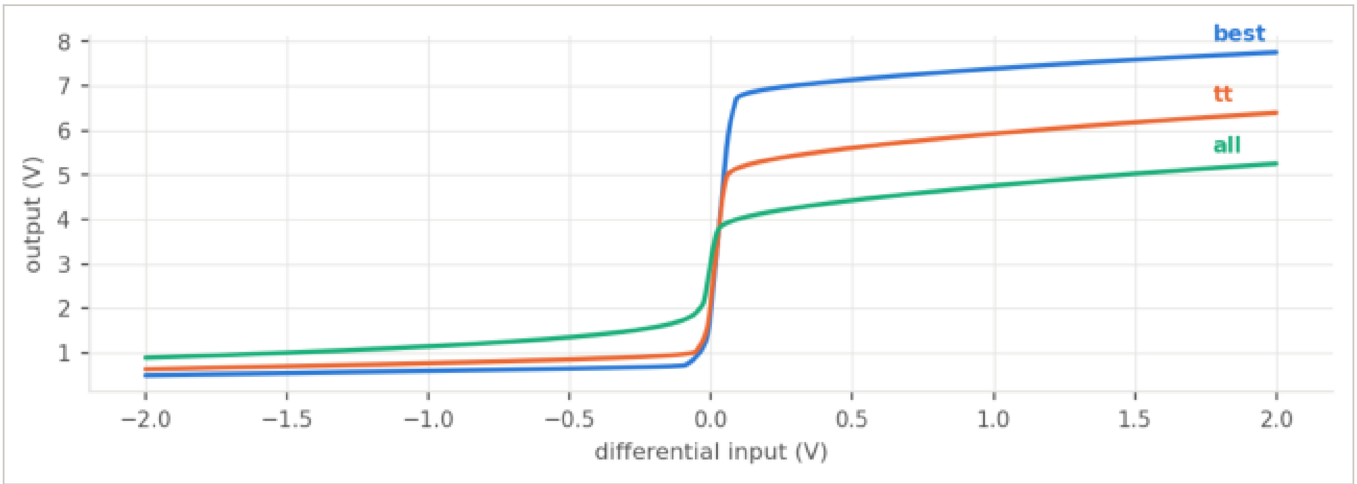


Gain and phase against frequency, three corners

OPAM2



Gain and phase against frequency, three corners



Output against differential input

Magnetic field

The wrapper divides the intrinsic width by 1 + (mu*b_scale*B)^2. Because level 1 only ever uses the product Kp*W/L, that is the same as dividing Kp, which is the same as dividing the mobility - so this models classical magnetoresistance, and Kn and channel resistance are one fact, not two. Contact resistance and overlap capacitance use the outer W and do not move with the field, which is where the physics leaves them.

What is NOT modelled: the Hall effect - a transverse voltage, and the effect an actual magnetometer is built from - and any shift of the threshold with field. If what the laboratory measures under field is one of those, fitting b_scale would be forcing the wrong mechanism.

At b_scale = 1 the effect is (mu*B)^2 = 2.1e-7 per tesla at the best corner: two parts in ten million, below the resolution of any measurement in this report. It also goes as mu squared, so the sensitivity changes by 31x between the best and the short corners. Classical magnetoresistance in this material is not a sensor.

The gain is blind to the field; the bandwidth is not

A uniform mobility change barely moves the gain. That is not luck, it is the topology: the gain is set by ratios of identical devices, and a change that scales all of them together leaves the ratios alone. Reduce the effective width by 95 % and the gain falls 0.6 dB - while the supply current falls 95 % and the bandwidth falls 82 %.

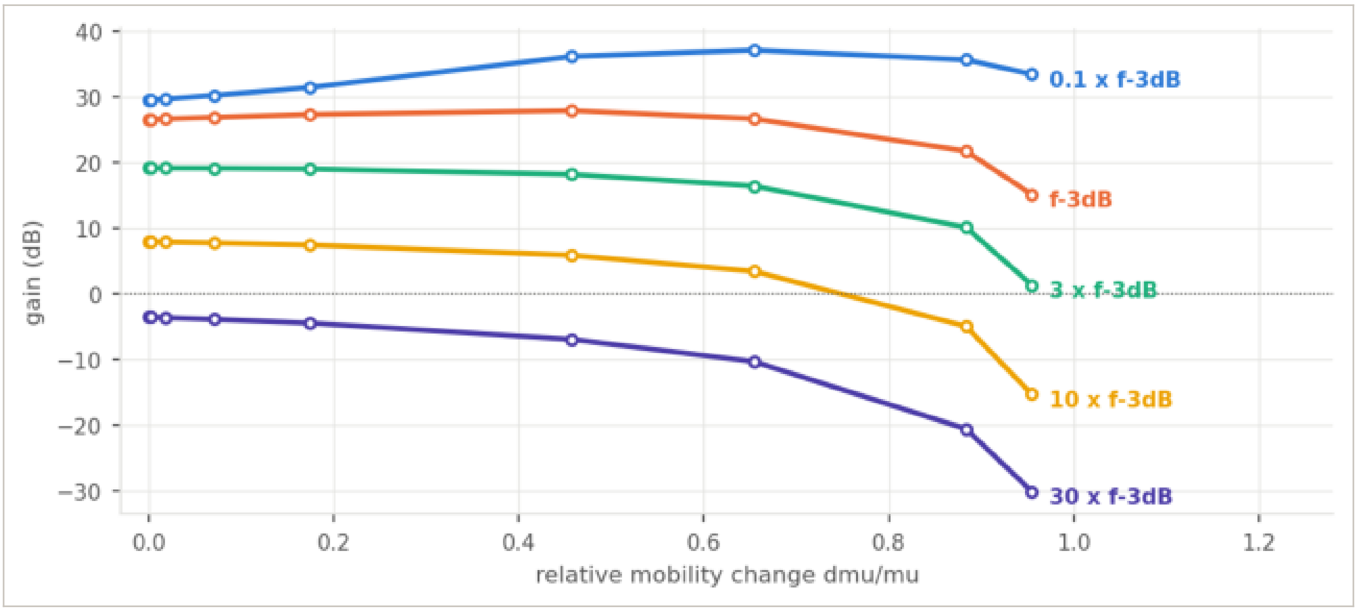
That points at the measurement to make. Park a probe tone where the response is steep, and a bandwidth shift becomes an amplitude change. On the corner it is worth about 2.5x the flat band; a decade past the corner, where the full roll-off is, OPAM gives 10.3 dB of output per unit of relative mobility change against 0.94 dB in the flat band - eleven times better.

Turned around, that is the number to hand the laboratory: for a tenth of a decibel of output change at one tesla, the measured effect would have to sit about 216 times above the classical one. If it does not, this amplifier is not the sensor - and the place to look is a topology where only one branch sees the field, so the ratios stop cancelling.

Gain against field, one curve per probe tone

Each curve is a tone held at a fixed frequency while the field moves the circuit underneath it. The lowest curve sits a decade below the corner, in the flat band, and barely moves - a mobility change scales every device together and the ratios that set the gain do not notice. The higher the tone sits on the roll-off, the more of the same change it reads: over the full sweep the flat-band tone loses 3.9 dB and the one at thirty times the corner loses 26.6 dB.

That is the measurement to build a sensor on, if the effect is ever large enough to measure: not the gain, which is blind to the field, but the gain read at a frequency where the response is steep.

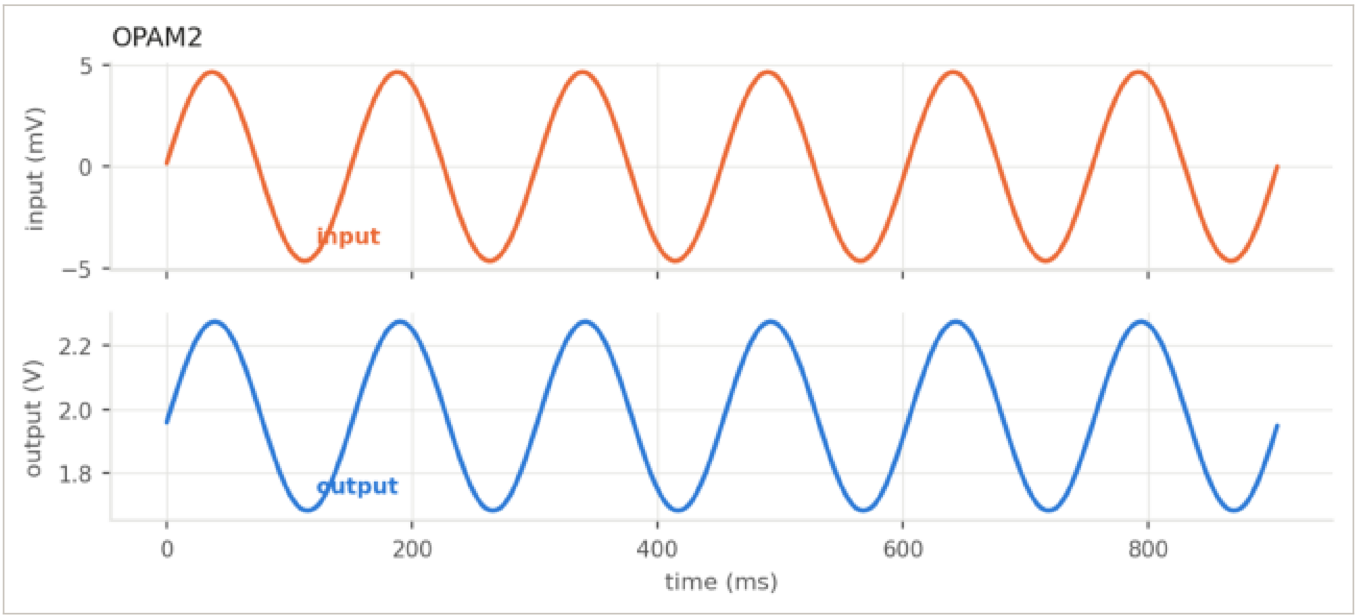


Gain against relative mobility change, one line per probe tone

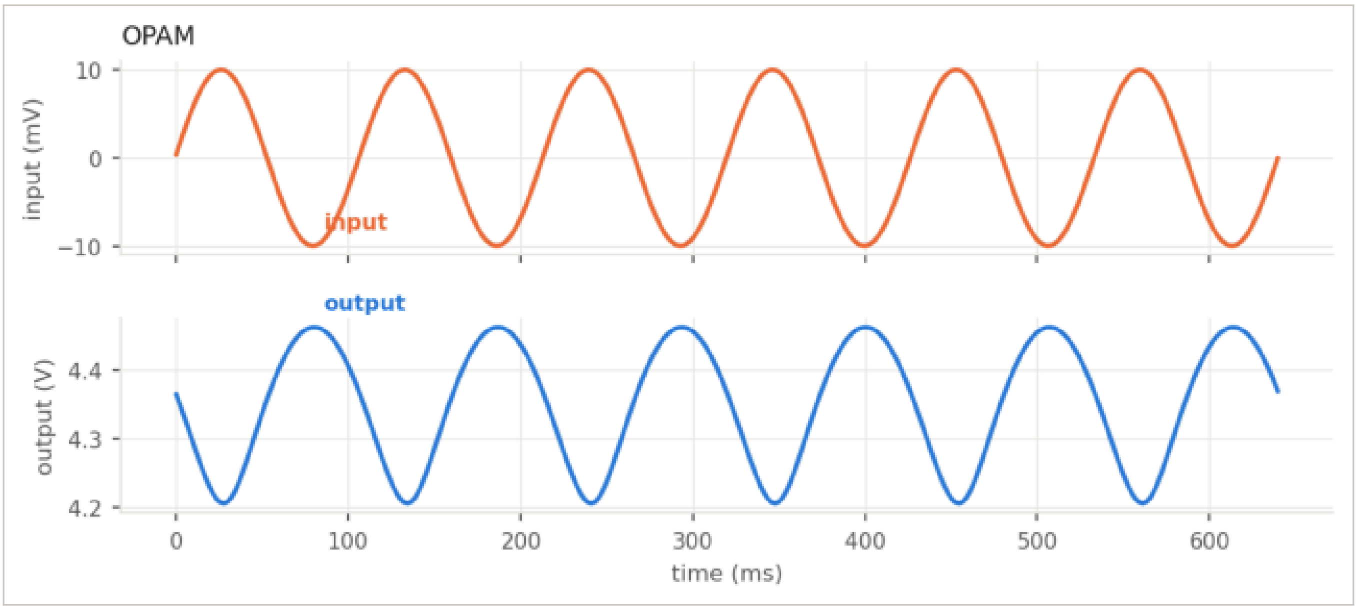
The signals in time

Input and output against time, at a tone a decade below the corner, with the amplitude chosen from the measured gain so the output lands near 0.6 V peak-to-peak and cannot clip. Input and output are drawn on two panels sharing the time axis rather than one pair of scales: the input is under ten millivolts and the output is hundreds, and on one scale the input would be a flat line.

This is also the check that the AC gain is not lying. The gain read off these waveforms and the gain read off the AC sweep at the same frequency agree to 0.02 dB on OPAM2. On OPAM they differ by 7.3 dB, and that difference is the subject of the page on what the AC analysis does not see.

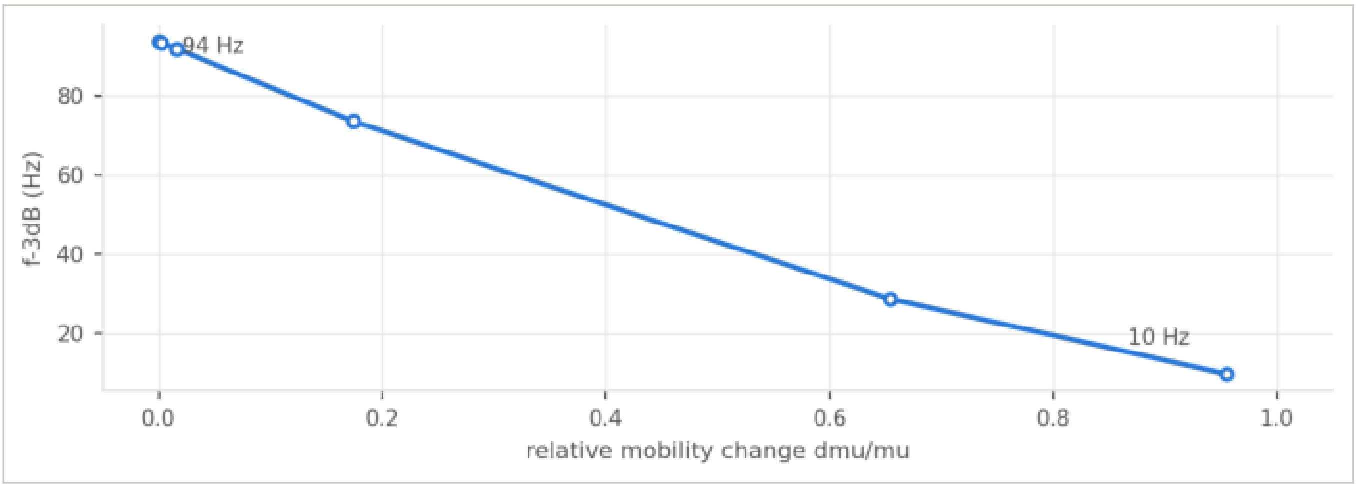


Input and output against time, best corner

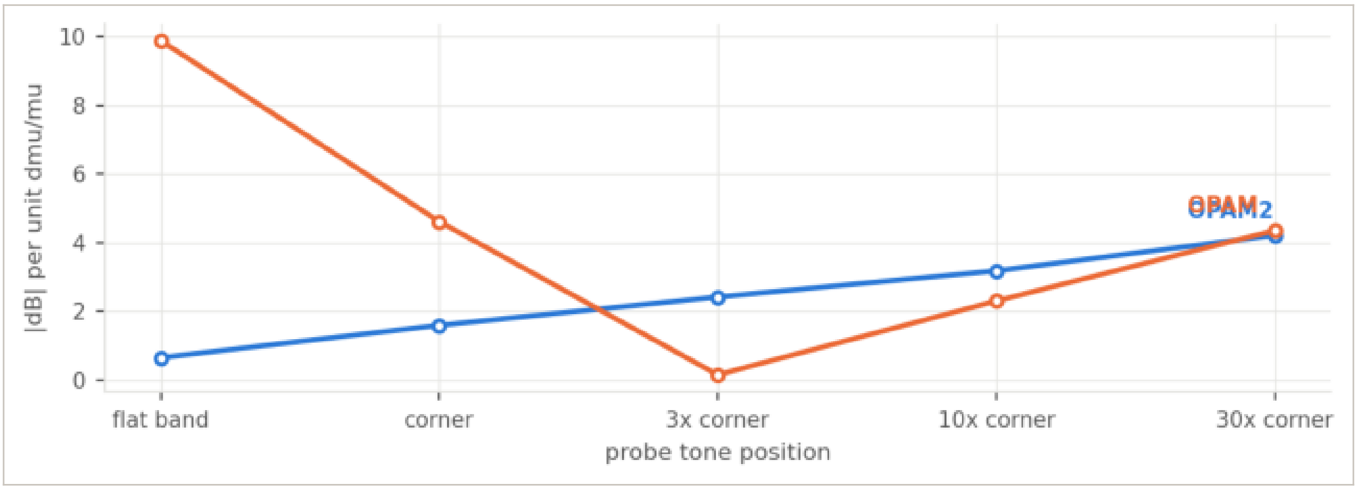


Input and output against time, best corner

The gain is blind to the field; the bandwidth is not



The -3 dB corner against relative mobility change



Sensitivity against where the probe tone sits

probe tone	f (Hz)	OPAM2	OPAM
flat band	7 / 9	-0.64	9.89
corner	66 / 94	-1.59	4.62
3x corner	199 / 281	-2.42	-0.14
10x corner	662 / 937	-3.18	-2.30
30x corner	1987 / 2811	-4.21	-4.36

dB per unit dmu/mu

What to trust, and what not to

Gain, DC operating point, swing, power trustworthy. The models reproduce measured output curves with a median simulated/measured of 1.01 over 18 devices, and every number here sits inside the validated box.

Bandwidth, phase margin, anything AC an order of magnitude. Cox is measured; the overlap that sets every capacitance was read off one GDS cell and never checked against a C-V.

The transient gain of OPAM more trustworthy than its AC gain, because the bootstrap node's DC level is set by a leakage the model puts at zero.

Magnetic field the classical term is right and negligible. Anything larger is a measurement this PDK has not made.

Level 1 runs about 20 % pessimistic on gm, because the measured saturation exponent is 2.31 rather than the square law's 2.